Final Project: AI-Powered Data Analysis and Prediction System



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Course project • Python • Machine Learning • Data Science • AI

1. Project Overview

FINAL PROJECT

AI-Powered Data Analysis and Prediction System

Course: Principles of Artificial Intelligence and Data Science

Project Type: Group Project

Recommended Group Size: 3 students

Programming Language: Python

Recommended Environment: Google Colab, Jupyter Notebook, or Visual Studio Code

Project Description

Students will design, develop, and evaluate an AI-powered system that uses data science techniques and machine learning algorithms to analyze a real-world problem.

The system must accept a dataset, perform data preprocessing, explore patterns, train or apply an appropriate machine learning algorithm, evaluate the results, and present meaningful insights through visualizations and a written report.

Students may select any approved application domain, provided the project demonstrates the core concepts of Artificial Intelligence and Data Science covered in the course.

Possible domains include education, healthcare research using public or anonymized data, business, agriculture, environmental monitoring, esports analytics, entertainment and multimedia, cybersecurity, and transportation.

The project must demonstrate both technical implementation and critical analysis. Simply running a prebuilt model without explaining its operation, evaluation, and limitations is insufficient.

General Objective

To develop a working AI and Data Science application that demonstrates how data can be transformed into useful predictions, discovered patterns, or actionable insights.

Specific Objectives

At the end of the project, students should be able to:

1. Identify and formulate a real-world problem suitable for data science.
2. Acquire or construct an appropriate dataset.
3. Clean, preprocess, and explore data.
4. Select appropriate AI or machine learning techniques.
5. Implement a working Python-based solution.
6. Evaluate model performance using appropriate metrics.
7. Visualize and interpret results.
8. Identify limitations, ethical concerns, and possible improvements.
9. Communicate findings through a technical report and oral presentation.

2. Choose a Project Topic

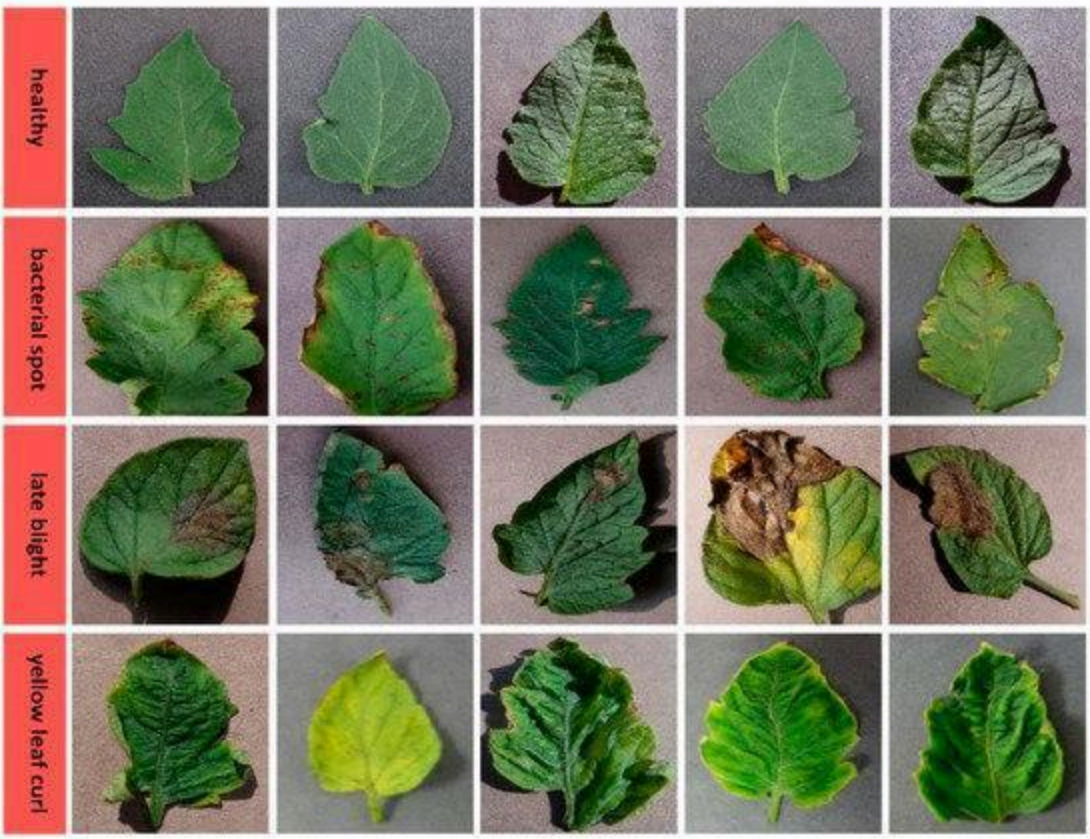
Students may propose their own topic or select from the following options.



Option A: Student Performance Analysis and Prediction

Predict an academic outcome using suitable historical data, or discover student performance patterns using clustering.

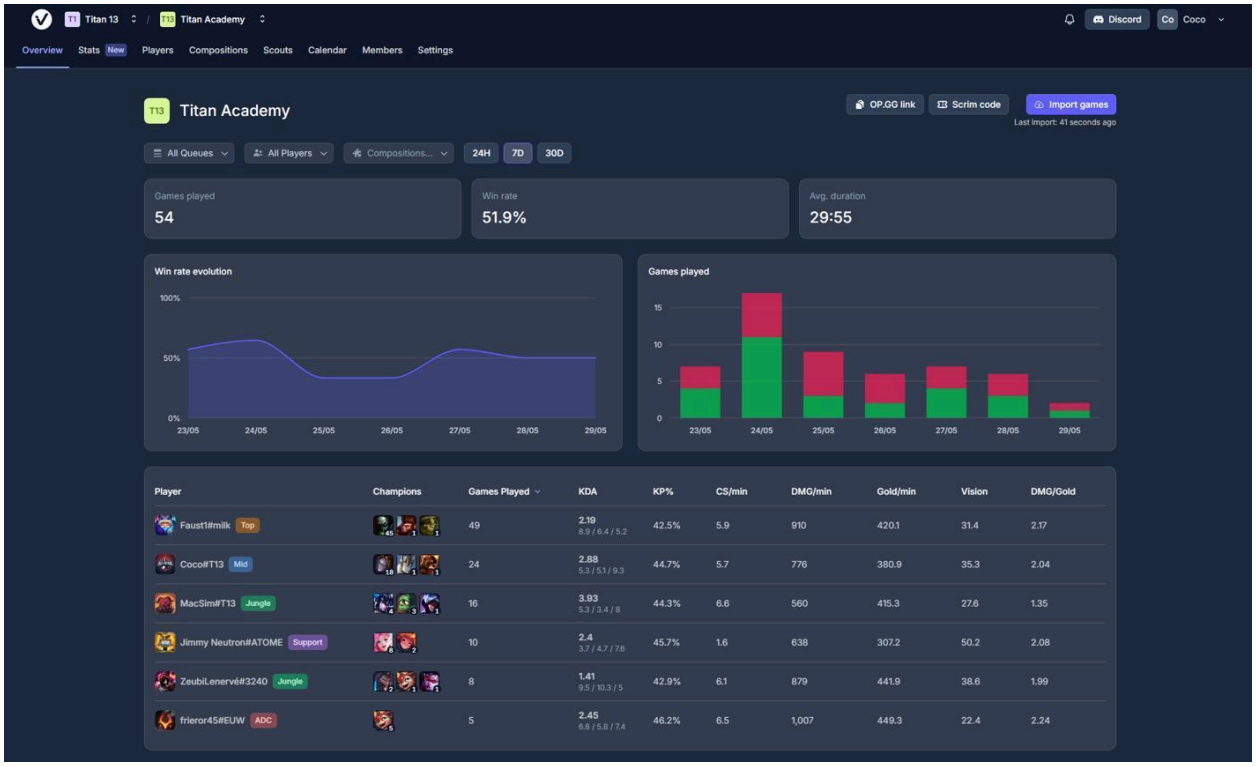
Possible techniques: Decision Tree, Logistic Regression, K-Means



Option B: AI for Agriculture

Classify plant leaf conditions from images or analyze agricultural measurements to discover patterns.

Possible techniques: CNN, image classification, clustering



Option C: Esports Performance Analytics

Analyze player statistics, predict an appropriate target outcome, or cluster players according to performance characteristics.

Possible techniques: Regression, classification, K-Means



Option D: Customer Segmentation and Sales Analytics

Identify customer groups or predict sales using a suitable business dataset.

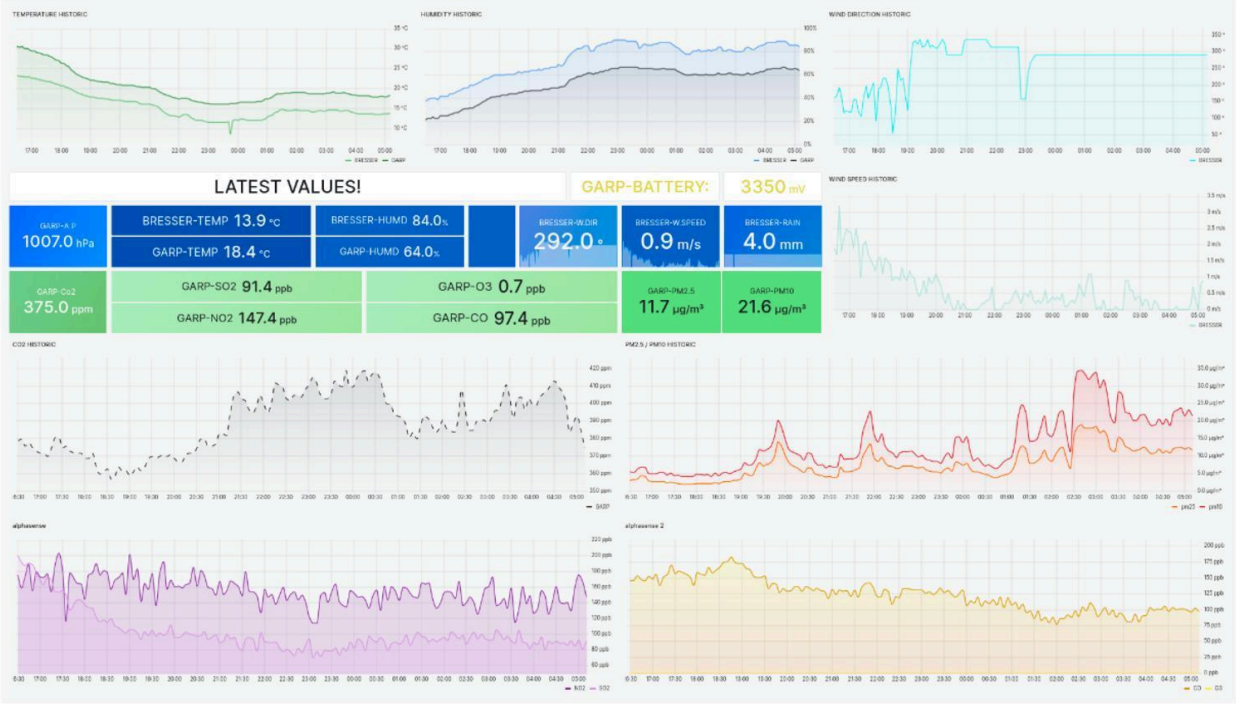
Possible techniques: K-Means, regression, time-series analysis



Option E: Multimedia and Entertainment Analytics

Analyze animation, game, movie, or audience datasets to predict outcomes or discover content patterns.

Possible techniques: Classification, clustering, recommendation methods



Option F: Environmental Data Intelligence

Analyze air quality, rainfall, temperature, waste, or other environmental measurements to identify trends or predict values.

Possible techniques: Regression, anomaly detection, clustering

Students proposing other topics must obtain instructor approval before development.

3. Required System Functionalities

Every project must include the following core functionalities, adapted to the selected problem.

1. Dataset Management

Load a dataset from a CSV file, Excel file, image collection, or another appropriate source. Display basic information about the data.

2. Data Preprocessing

Handle missing values, duplicates, inconsistent formats, outliers where appropriate, and categorical features. Explain preprocessing decisions.

3. Exploratory Data Analysis (EDA)

Generate descriptive statistics and at least three useful visualizations, such as histograms, scatter plots, correlation heatmaps, or category comparisons.

4. AI or Machine Learning Module

Implement at least one suitable algorithm. Depending on the project, it may perform classification, regression, clustering, dimensionality reduction, anomaly detection, or image analysis.

5. Model Evaluation

Evaluate the model using metrics appropriate to its task. Explain what the results mean and identify weaknesses.

6. Results and Visualization

Display predictions, discovered clusters, or analytical findings using readable tables, graphs, and summaries.

7. Ethics and Limitations

Discuss data privacy, potential bias, reliability, limitations, and appropriate use of the system.

4. Minimum Technical Requirements

Requirement	Minimum expectation
Dataset	Relevant and sufficiently documented
Data exploration	Descriptive statistics and at least 3 visualizations
Preprocessing	Appropriate cleaning and transformation
AI/ML	At least 1 implemented algorithm
Evaluation	Appropriate task-specific metrics
Experimentation	At least 1 meaningful comparison or parameter experiment
Results	Interpreted outputs and visualizations
Code	Runnable, organized, and commented
Ethics	Written discussion of limitations and responsible use

Dataset size: There is no universal minimum number of rows. Students must justify that their dataset is sufficient for the intended analysis. Small demonstration datasets may be acceptable for selected tasks, but real-world claims require adequate evidence.

For supervised learning, separate training and testing data appropriately. Do not use the test set to repeatedly tune the model. For unsupervised learning, justify the selected number of clusters and evaluate the resulting groupings where feasible.

5. Required Documentation

FINAL PROJECT DOCUMENTATION

Chapter 1: Introduction

Include:

- Background of the study
- Problem statement
- General and specific objectives
- Scope and limitations

- Significance of the project

Chapter 2: Review of Related Systems and Concepts

Discuss:

- Relevant AI and Data Science concepts
- Related studies or existing systems
- Algorithms selected for the project
- Identified limitations or opportunities for improvement

Use credible sources and APA 7 citations where applicable.

Chapter 3: Methodology

Include:

- Research or development design
- Dataset source and description
- Data collection procedures
- Data preprocessing steps
- Feature selection
- Algorithm selection and justification
- System architecture or workflow
- Experimental design
- Evaluation metrics
- Ethical considerations

Chapter 4: Results and Discussion

Include:

- Dataset description
- Exploratory data analysis
- Preprocessing results
- Model configuration

- Evaluation results
- Tables and visualizations
- Interpretation of findings
- Comparison of experimental results
- Limitations and unexpected results

Chapter 5: Conclusion and Recommendations

Include:

- Summary of the completed system
- Findings based on the results
- Achievement of objectives
- Limitations
- Recommended future improvements

References

List all sources cited in the documentation.

Appendices

Include:

- Source code or repository link
- Dataset description or permitted sample
- Additional screenshots
- Test cases
- User guide
- Additional tables and graphs

6. Development and Submission Deliverables

Students must submit the following:

Submission checklist

Project proposal

Problem, objectives, dataset, proposed algorithm, and expected output.

Runnable source code

Python scripts or notebook with setup instructions and required dependencies.

Dataset and data dictionary

Dataset source, feature descriptions, and applicable license or usage permissions.

Technical documentation

Complete report using the required chapter structure.

Working demonstration

A notebook, command-line application, or interface that demonstrates the core functionalities.

Presentation

Slides explaining the problem, method, results, limitations, and conclusions.

7. Suggested 8-Week Project Timeline

Week	Activity	Expected output
1	Topic selection and problem definition	Approved proposal
2	Dataset acquisition and exploration	Dataset and EDA
3	Data cleaning and preprocessing	Prepared dataset
4	Algorithm implementation	Initial working model
5	Experimentation and evaluation	Metrics and comparison
6	Visualization and application integration	Demonstrable system

7	Testing, interpretation, and documentation	Draft report
8	Final presentation and defense	Complete submission

The timeline can be compressed or expanded according to the remaining weeks of the semester.

8. Grading Rubric

Total: 100 points.

Criteria	**Weight**	**Excellent performance**
Problem definition and objectives	10	Clear, relevant, measurable problem
Dataset and preprocessing	15	Appropriate data handling with justified decisions
Algorithm implementation	20	Correct, functional, and well-explained implementation
Evaluation and experimentation	15	Suitable metrics and meaningful comparisons
Visualization and interpretation	10	Clear graphs and evidence-based insights
System functionality and code quality	10	Runnable, organized, documented solution
Technical report	10	Complete, coherent, properly referenced
Presentation and defense	5	Clear explanation and competent responses
Ethics and limitations	5	Thoughtful, realistic analysis
Total	100	

Performance Guide

- 90–100: Excellent — technically sound, well-evaluated, and insightful.
- 85–89: Very Good — complete with minor weaknesses.
- 80–84: Good — functional but needs improvements.
- 75–79: Satisfactory — basic requirements achieved.
- Below 75: Needs substantial improvement, subject to institutional grading policies.

9. Final Presentation and Defense

Each group should deliver a 10–15-minute presentation followed by a 5-minute question-and-answer session.

The presentation must explain:

1. What problem does the project address?
2. Why is AI or Data Science appropriate?
3. Where did the dataset come from?
4. How was the data prepared?
5. Why was the algorithm selected?
6. How does the system work?
7. What do the evaluation results demonstrate?
8. What are the limitations?
9. What ethical concerns need to be addressed?
10. What improvements would the group implement next?

Every member must be able to explain the group's methodology and demonstrate a meaningful understanding of the code.

10. Academic Integrity and Responsible AI Use

Students may use AI tools to assist with brainstorming, debugging, documentation, and understanding algorithms, subject to instructor policies.

However:

- Students must understand and be able to explain their submitted code.
- Results and experimental findings must not be fabricated.
- Sources, datasets, and AI assistance must be disclosed as required by the instructor.
- Student work must comply with dataset licenses, privacy requirements, and institutional research policies.
- Copying another group's project without proper authorization or acknowledgment is not acceptable.